Sahaj Bhadja

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Summary

Software Engineer with 2+ years troubleshooting and fixing production systems in **Java** and Python, performing **root-cause analysis** from logs, shipping tested fixes with Mockito and PyTest, and running **Snyk** and OWASP scans, using Claude Code daily.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Diagnosed **production performance** issues on a multi-tenant SQL Server and Azure SQL platform, tracing slow dashboards to query and index patterns and cutting latency from 1.2s to 350ms for 500+ users.
- Restructured failing async job processing with explicit job-state transitions and failure isolation to stop **cascading failures**, tripling throughput across 1K+ jobs and sustaining 99%+ reliability over 5K+ monthly integrations.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Resolved **OCR reliability** and timeout failures in a production grading platform by adding fallback routing across external model providers, keeping 10K+ daily requests stable for 150+ instructors at peak load.
- Built secure PDF, DOCX, and ZIP file ingestion with concurrent batch processing of 1K+ files and fault-tolerant **error handling** over REST APIs and MySQL, shipping fixes through GitHub Actions CI/CD.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Troubleshot** and optimized large-scale API integrations in object-oriented Singleton and Factory modules, improving response times 30% and cutting technical debt 20% across Windows, Linux, and macOS builds.
- Wrote 20+ unit and integration test suites with **Mockito** and PyTest in nightly Jenkins pipelines that built, installed, and validated executables, emailing failure logs and coverage reports for defect triage.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Ran **Snyk** and OWASP Dependency-Check vulnerability scans with linting and static checks in Jenkins CI/CD, gating staging and production releases behind manual approval at 95%+ build validation.
- Debugged production errors with Sentry, Azure Monitor, and Kibana logs across eight Docker-packaged REST microservices with JWT authentication, synchronizing 2M+ records across multiple locations in real time via WebSockets.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **Java Spring Boot** REST APIs for bookings, pricing, and payments with JSON payloads and JWT auth, integrating existing AWS-hosted microservices and sustaining sub-200ms latency across 50K+ records.
- Kept payment status consistent through idempotent **webhooks** for refunds, disputes, and chargebacks, cutting manual payment handling by 70%, and presented the end-to-end 12+ integration-point architecture to the client.

Skills

Languages & Java: Java, JDK, Spring Boot, JUnit, Mockito, Maven, Python, SQL, JavaScript, HTML
APIs & Formats: REST APIs, HTTP(S), JSON, XML, YAML, Webhooks, JWT, TCP/IP
Databases: SQL Server, MySQL, Oracle, PostgreSQL, Azure SQL, Query Optimization
Cloud & CI/CD: AWS, Azure, Docker, Kubernetes, Jenkins, GitHub, GitHub Actions, Linux, Windows
Security & Debugging: Snyk, OWASP Dependency-Check, Static Analysis, Root Cause Analysis, Log Analysis, Sentry, Kibana
AI & Practices: Claude Code, AI-Assisted Debugging, Technical Documentation, Agile

Projects

B2B File Transfer Gateway | *Java, Spring Boot, JMS, SFTP, JUnit, Docker*

- Designed a Java **file transfer gateway** that receives partner payloads over SFTP and HTTPS, queues them on JMS with retries and dead-letter handling, verifies signatures, and logs traceable delivery receipts.

Chat2SQL Code | *Java 17, Maven, TypeScript, Oracle, SQL Server, Docker*

- Navigated and extended a large open-source **Java** 17 and TypeScript codebase with Maven and Docker builds, supporting 10+ databases including Oracle and SQL Server and AI-generated SQL at 82% accuracy.

Comprehensive Analysis of Blockchain for User Authentication Paper | *Authentication, Security*

- Co-authored an IEEE paper analyzing authentication **security** threats including identity theft, replay, man-in-the-middle, and phishing attacks, comparing centralized identity stores with smart-contract verification for users and devices.